ПРВИ КОЛОКВИЈУМ – 19.04.2019

ЗАДАТАК:

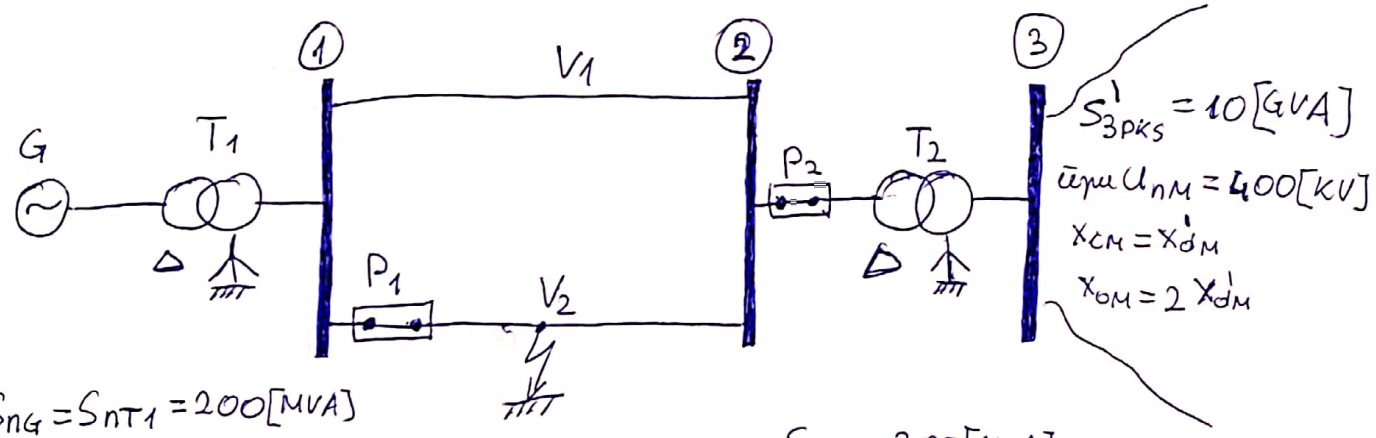
На слици је приказан једноставан ЕЕС. За случај 1ПКС фазе А на средини преносног вода V_2, потребно је одредити:

а) струју 1ПКС

б) струје по фазама кроз прекидаче P_1 и P_2

в) да ли су добро подешени прекидачи P_1 и P_2 ако је струја реаговања оба прекидача $I_{P1r} = I_{P2r} = 0.6$ [kA] и ако је њихова улога да искључе само фазу која је погођена кваром

Напон на месту квара пре настанка квара је износио $U_{fr} = 220/\sqrt{3}$ [kV].



$S_{nG} = S_{nT1} = 200$ [MVA]

$U_{nG} = 15.75$ [kV]

$m_{T1} = \frac{15.75}{231} \frac{[kV]}{[kV]}$

$X'_{dG\%} = X_{iG\%} = 18\%$

$X_{T1\%} = 12\%$

$U_{nV} = 220$ [kV]

$x_V = 0.4 \left[\frac{\Omega}{km}\right]$

$x_{0V} = 1.2 \left[\frac{\Omega}{km}\right]$

$L_V = 100$ [km]

$S_{nT2} = 300$ [MVA]

$m_{T2} = \frac{220}{400} \frac{[kV]}{[kV]}$

$X_{T2\%} = 11\%$

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РЈЕШЕЊЕ:

- Вриједности параметара система сведени на напонски ниво 220 [kV] (на тој страни десет квар)

• генератор:

$$X'_{dG} = X_{iG} = \frac{X'_{dG\%}}{100} \cdot \frac{U_{nG}^2}{S_{nG}} \cdot \frac{1}{m_{T1}^2} = \frac{18}{100} \cdot \frac{15.75^2}{200} \cdot \frac{231^2}{15.75^2}$$

$$X'_{dG} = X_{iG} = 48.0249\,[\Omega]$$

• Тф-а

$$X_{T1} = \frac{X_{T1\%}}{100} \cdot \frac{U_{nT1}^2}{S_{nT1}} = \frac{12}{100} \cdot \frac{231^2}{200} = 32.0166\,[\Omega]$$

$$X_{T2} = \frac{X_{T2\%}}{100} \cdot \frac{U_{nT2}^2}{S_{nT2}} = \frac{11}{100} \cdot \frac{220^2}{300} = 17.7467\,[\Omega]$$

• вод:

$$X_{dv} = X_{iv} = x_v L_v = 0.4 \cdot 100 = 40\,[\Omega]$$

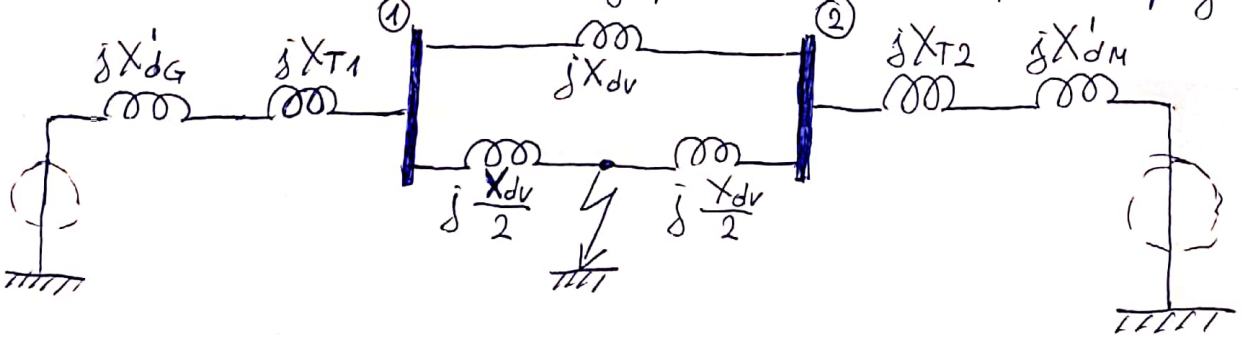
$$X_{ov} = x_{ov} L_v = 1.2 \cdot 100 = 120\,[\Omega]$$

• јака мрежа:

$$X'_{dM} = \frac{U_{nM}^2}{S'_{3PKS}} \cdot m_{T2}^2 = \frac{400^2}{10\,000} \cdot \frac{220^2}{400^2} = 4.8400\,[\Omega]$$

$$X_{oM} = 2X'_{dM} = 2 \cdot 4.84 = 9.6800\,[\Omega]$$

- Еквивалентна шема директног / инверзног редослиједа



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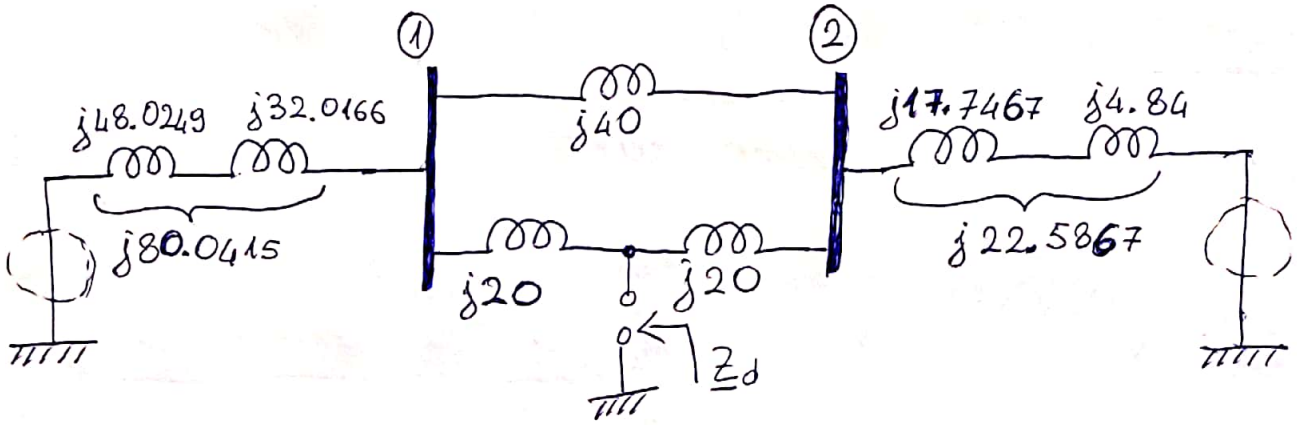
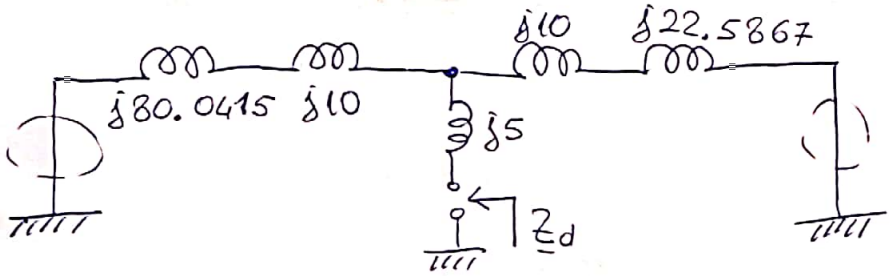


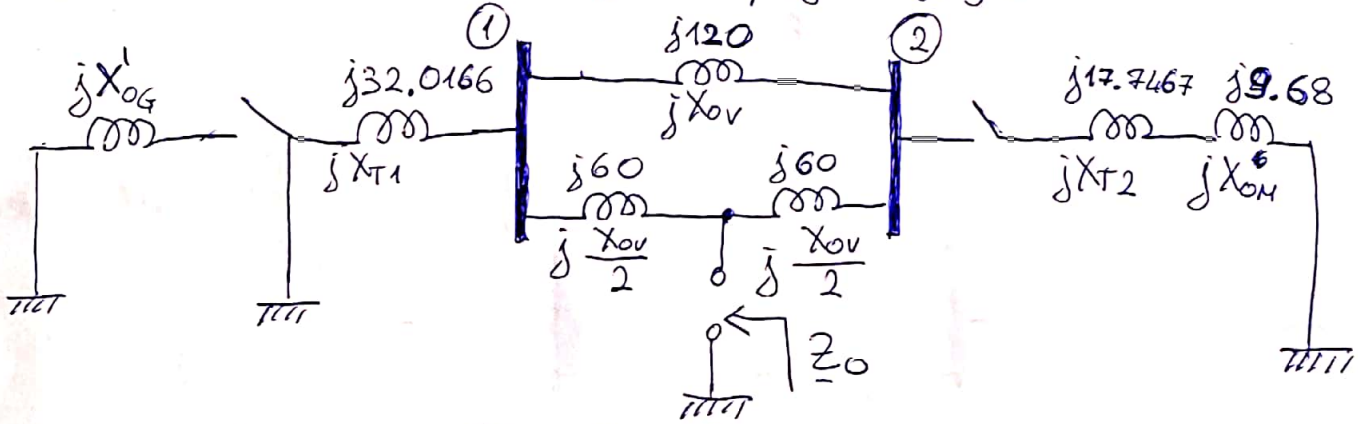
- Потребно је трансформисати троугао између сабирнице број 1 и сабирнице број 2 у звијезду

$$\underline{Z}' = \frac{j20 \cdot j40}{j(20+20+40)} = j10[\Omega] \quad ; \quad \underline{Z}'' = \frac{j20 \cdot j20}{j(20+20+40)} = j5[\Omega]$$



$$\underline{Z}_d = \underline{Z}_i = \frac{j(80.0415+10) \cdot j(10+22.5867)}{j(80.0415+10+10+22.5867)} + j5 = j28.9272[\Omega]$$

- Еквивалентна шема нултог редослиједа



• трансформација $\Delta \rightarrow Y$

$$\underline{Z}' = \frac{j60 \cdot j120}{j(60+60+120)} = j30[\Omega] \quad ; \quad \underline{Z}'' = \frac{j60 \cdot j60}{j(60+60+120)} = j15[\Omega]$$

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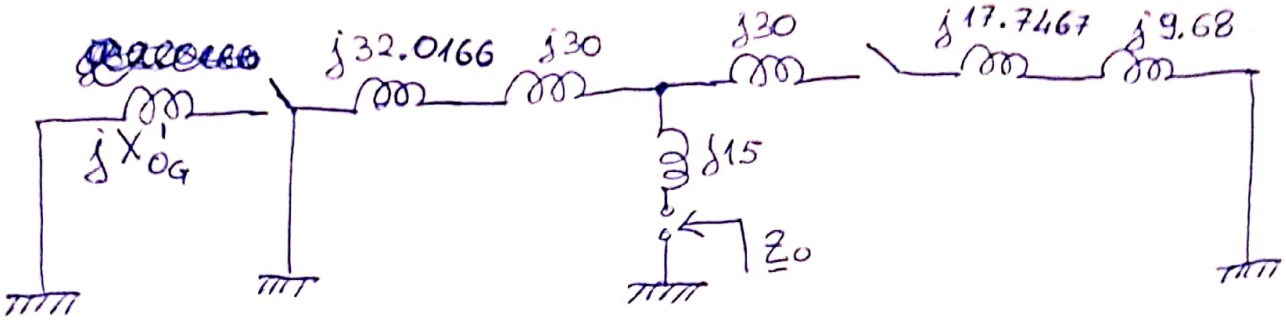


$$\underline{Z}_0 = j(32.0166 + 30 + 15) = j\,77.0166 \ [\Omega]$$

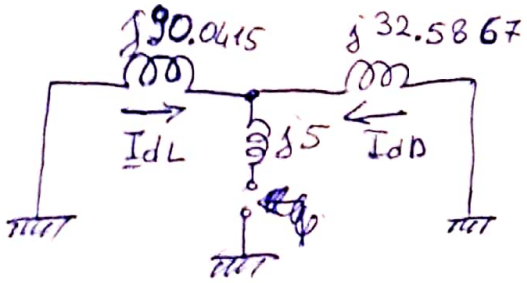
1PKS

$$\underline{I}_d = \underline{I}_i = \underline{I}_0 = \frac{\underline{U}_{fr}}{\underline{Z}_d + \underline{Z}_i + \underline{Z}_0} = \frac{220/\sqrt{3}}{2 \cdot j28.9272 + j77.0166}$$

$$\underline{I}_d = \underline{I}_i = \underline{I}_0 = -j0.9418 \ [kA]$$

a) $\underline{I}_{1PKS} = 3\,\underline{I}_0 = 3 \cdot (-j0.9418) \Rightarrow \boxed{\underline{I}_{1PKS} = -j2.8253 \ [kA]}$

b) - Директна и инверзна компонента



$$\underline{I}_{dL} = \frac{j32.5867}{j(90.0415 + 32.5867)} \cdot \underline{I}_d$$

$$\underline{I}_{dL} = \frac{32.5867}{122.6282} \cdot (-j0.9418) = -j0.2503 \ [kA]$$

$$\underline{I}_{iL} = \underline{I}_{dL} = -j0.2503 \ [kA]$$

$$\underline{I}_{dD} = \frac{j90.0415}{j(90.0415 + 32.5867)} \cdot \underline{I}_d = \frac{90.0415}{122.6282} \cdot (-j0.9418) = -j0.6915 \ [kA]$$

$$\underline{I}_{iD} = \underline{I}_{dD} = -j0.6915 \ [kA]$$

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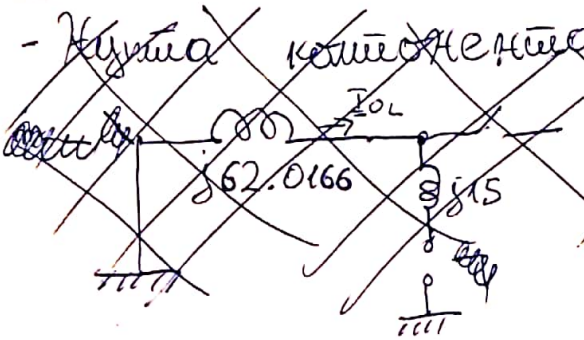
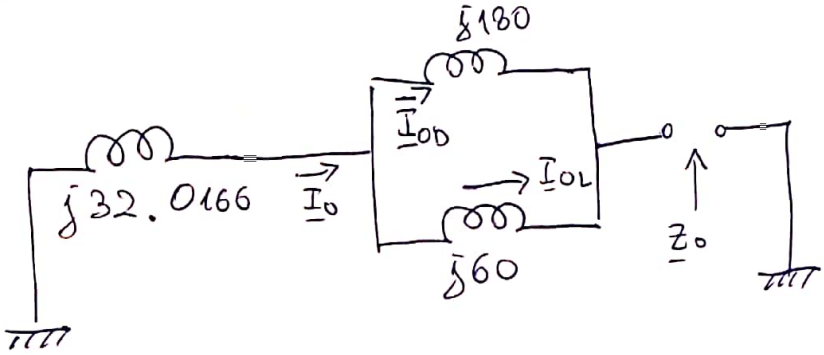
- ~~Нулта компонента~~



$$\Rightarrow \underline{I}_{0L} = \underline{I}_0 = -j0.9418\,[kA]$$

$$\underline{I}_{0D} = 0$$

- Нулта компонента



$$\underline{I}_{0D} = \frac{j60}{j(60+180)} \cdot \underline{I}_0 = \frac{60}{240} \cdot (-j\,0.9418) = -j0.2354\,[kA]$$

$$\underline{I}_{0L} = \frac{j180}{j(60+180)} \cdot \underline{I}_0 = \frac{180}{240} \cdot (-j0.9418) = -j0.7063\,[kA]$$

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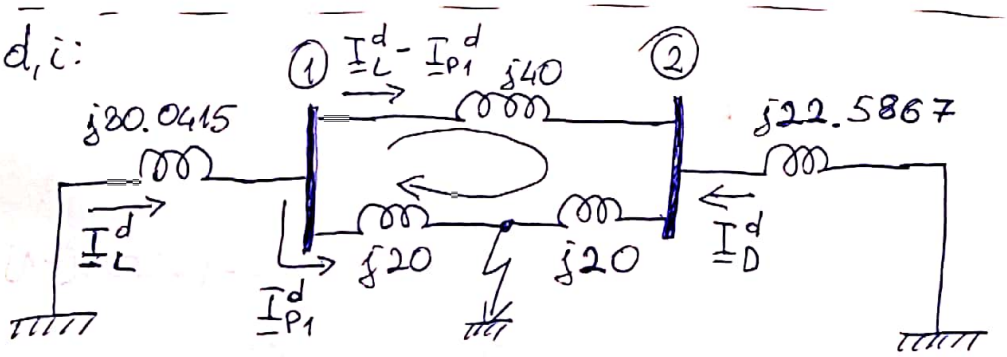
d, i:



$$\underline{I}_{P1} = \underline{I}_{P1}^d + \underline{I}_{P1}^i + \underline{I}_{P1}^0 \quad ; \quad \underline{I}_{P1}^d = \underline{I}_{P1}^i$$

$$(\underline{I}_L^d - \underline{I}_{P1}^d)\cdot j40 + (\underline{I}_L^d - \underline{I}_{P1}^d + \underline{I}_D^d)\cdot j20 = \underline{I}_{P1}^d \cdot j20$$

$$\underline{I}_L^d (j40 + j20) + \underline{I}_D^d \cdot j20 = \underline{I}_{P1}^d (j20 + j20 + j40)$$

$$j60\,\underline{I}_L^d + j20\,\underline{I}_D^d = j80\,\underline{I}_{P1}^d \;/ : j80$$

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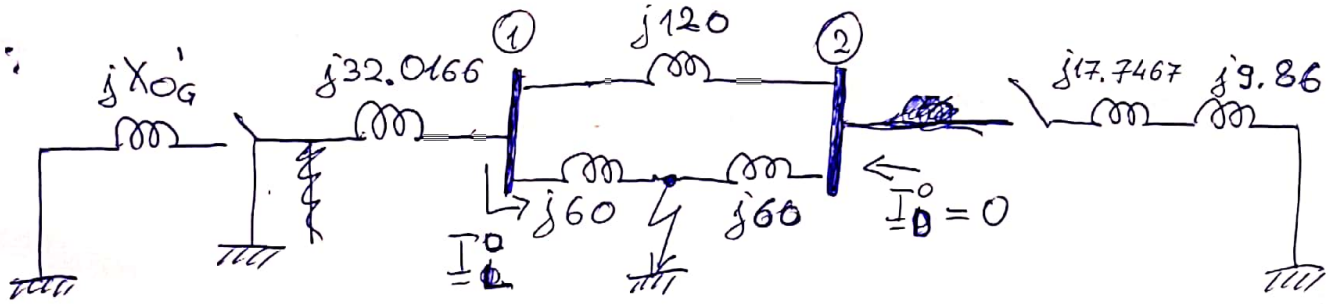
$$\underline{I}_{P1}^{d} = 0.75\,\underline{I}_{L}^{d} + 0.25\,\underline{I}_{D}^{d}$$

$$\underline{I}_{L}^{d} = -j0.2503\ [kA] \quad ; \quad \underline{I}_{D}^{d} = -j0.6915\ [kA]$$

$$\underline{I}_{P1}^{d} = 0.75\cdot(-j0.2503) + 0.25\cdot(-j0.6915)$$

$$\Rightarrow \underline{I}_{P1}^{d} = \underline{I}_{P1}^{i} = -j0.3606\ [kA]$$

O:



$$\underline{I}_{P1}^{o} = \underline{I}_{L}^{o} = -j0.7063\ [kA]$$

- Струје по фазама кроз P1 су:

$$\begin{bmatrix} \underline{I}_{P1}^{A} \\ \underline{I}_{P1}^{B} \\ \underline{I}_{P1}^{C} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ \underline{a}^2 & \underline{a} & 1 \\ \underline{a} & \underline{a}^2 & 1 \end{bmatrix} \begin{bmatrix} \underline{I}_{P1}^{d} \\ \underline{I}_{P1}^{i} \\ \underline{I}_{P1}^{o} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ \underline{a}^2 & \underline{a} & 1 \\ \underline{a} & \underline{a}^2 & 1 \end{bmatrix} \begin{bmatrix} -j0.3606 \\ -j0.3606 \\ -j0.7063 \end{bmatrix}$$

$$\underline{I}_{P1}^{A} = -j0.3606 - j0.3606 - j0.7063 = -j1.4275\ [kA]$$

$$\underline{I}_{P1}^{B} = -j0.3606\,a^2 - j0.3606\,a - j0.7063 =$$

$$= -j0.3606\left(-0.5 - j\frac{\sqrt{3}}{2}\right) - j0.3606\left(-0.5 + j\frac{\sqrt{3}}{2}\right) - j0.7063 =$$

$$= -j0.3606\left(-0.5 \cancel{- j\frac{\sqrt{3}}{2}} - 0.5 \cancel{+ j\frac{\sqrt{3}}{2}}\right) - j0.7063 = -j0.3457\ [kA]$$

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$$\underline{I}_{P1}^{C} = -j0.3606\,\underline{a} - j0.3606\,\underline{a}^2 - j0.7063 =$$

$$= -j0.3606\left(-0.5 + \cancel{j\frac{\sqrt{3}}{2}} - 0.5 - \cancel{j\frac{\sqrt{3}}{2}}\right) - j0.7063 = -j0.3457\,[kA]$$

$$\boxed{\begin{bmatrix} \underline{I}_{P1}^{A} \\ \underline{I}_{P1}^{B} \\ \underline{I}_{P1}^{C} \end{bmatrix} = \begin{bmatrix} -j1.4275 \\ -j0.3457 \\ -j0.3457 \end{bmatrix} [kA]}$$

- Струје по фазама кроз P_2 су:

$$\begin{bmatrix} \underline{I}_{P2}^{A} \\ \underline{I}_{P2}^{B} \\ \underline{I}_{P2}^{C} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ \underline{a}^2 & \underline{a} & 1 \\ \underline{a} & \underline{a}^2 & 1 \end{bmatrix} \begin{bmatrix} \underline{I}_{P2}^{d} \\ \underline{I}_{P2}^{i} \\ \underline{I}_{P2}^{o} \end{bmatrix}$$

$$\underline{I}_{P2}^{d} = \underline{I}_{P2}^{i} = I_{D}^{d} = -j0.6915\,[kA]$$

$$\underline{I}_{P2}^{o} = \underline{I}_{D}^{o} = 0$$

$$\begin{bmatrix} \underline{I}_{P2}^{A} \\ \underline{I}_{P2}^{B} \\ \underline{I}_{P2}^{C} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ \underline{a}^2 & \underline{a} & 1 \\ \underline{a} & \underline{a}^2 & 1 \end{bmatrix} \begin{bmatrix} -j0.6915 \\ -j0.6915 \\ 0 \end{bmatrix}$$

$$\underline{I}_{P2}^{A} = -j0.6915 - j0.6915 + 0 = -j1.3830\,[kA]$$

$$\underline{I}_{P2}^{B} = -j0.6915\,\underline{a}^2 - j0.6915\,\underline{a} + 0 =$$

$$= -j0.6915\left(-0.5 - \cancel{j\frac{\sqrt{3}}{2}} - 0.5 + \cancel{j\frac{\sqrt{3}}{2}}\right) = j0.6915\,[kA]$$

$$\underline{I}_{P2}^{C} = -j0.6915\,\underline{a} - j0.6915\,\underline{a}^2 =$$

$$= -j0.6915\left(-0.5 + \cancel{j\frac{\sqrt{3}}{2}} - 0.5 - \cancel{j\frac{\sqrt{3}}{2}}\right) = j0.6915\,[kA]$$

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$$\boxed{\begin{bmatrix} \underline{I}_{P2}^{A} \\ \underline{I}_{P2}^{B} \\ \underline{I}_{P2}^{C} \end{bmatrix} = \begin{bmatrix} -j1.3830 \\ -j0.6915 \\ -j0.6915 \end{bmatrix} [kA]}$$

c)

$$\left.\begin{aligned} I_{P1}^{A} &= 1.4275\,[kA] > 0.6\,[kA] = I_{P1r} \\ I_{P2}^{A} &= 1.3830\,[kA] > 0.6\,[kA] = I_{P2r} \end{aligned}\right\} \Rightarrow \text{треба да се реагира}$$

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